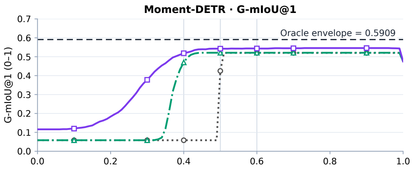
SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E6) + GEQ

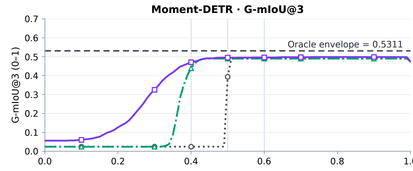
Standard-trained, fixed Val-selected best checkpoints → SoccerGMR Standard Test (1,036 queries)

Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]

Moment-DETR · G-mIoU@1



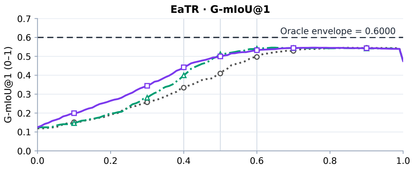
Moment-DETR · G-mIoU@3



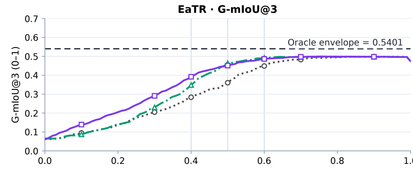
GMR baseline | mAP 6.68% · GMR + GEQ Full | mAP 5.77% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 14.97%

ΔmAP vs GMR: +8.29 pp (vs GEQ: +9.20 pp) · ΔG1 mean vs GMR (.4/.5/.6): +20.05 pp · ΔG3 mean vs GMR (.4/.5/.6): +18.47 pp

EaTR · G-mIoU@1



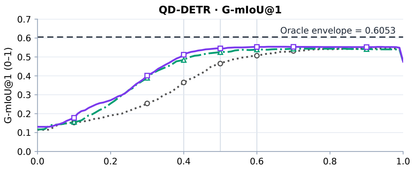
EaTR · G-mIoU@3



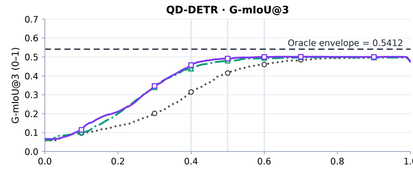
GMR baseline | mAP 16.49% · GMR + GEQ Full | mAP 17.48% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 15.54%

ΔmAP vs GMR: -0.95 pp (vs GEQ: -1.94 pp) · ΔG1 mean vs GMR (.4/.5/.6): +7.81 pp · ΔG3 mean vs GMR (.4/.5/.6): +7.87 pp

QD-DETR · G-mIoU@1



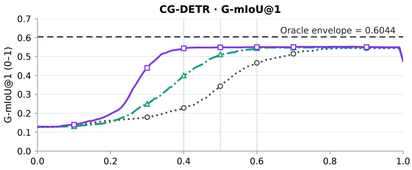
QD-DETR · G-mIoU@3



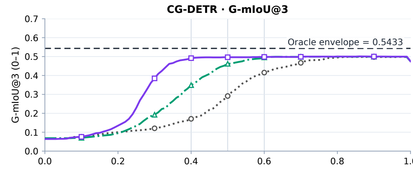
GMR baseline | mAP 14.62% · GMR + GEQ Full | mAP 14.83% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 18.27%

ΔmAP vs GMR: +3.65 pp (vs GEQ: +3.44 pp) · ΔG1 mean vs GMR (.4/.5/.6): +9.11 pp · ΔG3 mean vs GMR (.4/.5/.6): +8.63 pp

CG-DETR · G-mIoU@1



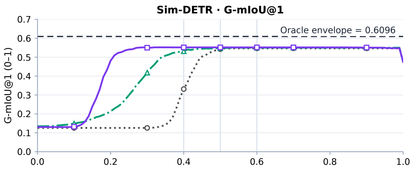
CG-DETR · G-mIoU@3



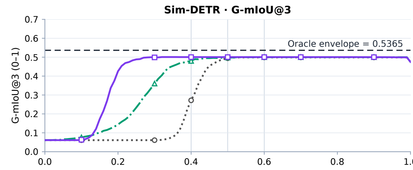
GMR baseline | mAP 18.65% · GMR + GEQ Full | mAP 18.71% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 17.66%

ΔmAP vs GMR: -0.99 pp (vs GEQ: -1.05 pp) · ΔG1 mean vs GMR (.4/.5/.6): +20.14 pp · ΔG3 mean vs GMR (.4/.5/.6): +20.30 pp

Sim-DETR · G-mIoU@1



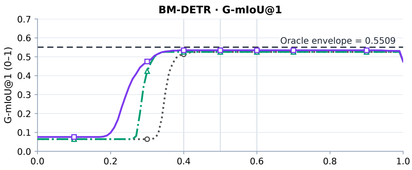
Sim-DETR · G-mIoU@3



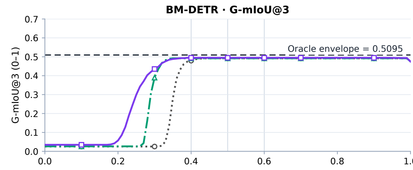
GMR baseline | mAP 20.73% · GMR + GEQ Full | mAP 22.26% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 21.14%

ΔmAP vs GMR: +0.41 pp (vs GEQ: -1.12 pp) · ΔG1 mean vs GMR (.4/.5/.6): +7.83 pp · ΔG3 mean vs GMR (.4/.5/.6): +7.88 pp

BM-DETR · G-mIoU@1



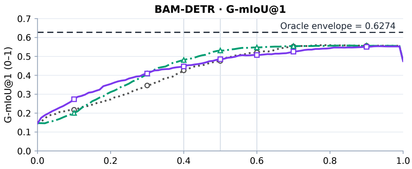
BM-DETR · G-mIoU@3



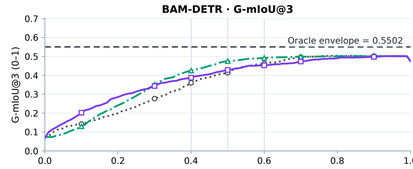
GMR baseline | mAP 4.00% · GMR + GEQ Full | mAP 4.27% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 6.17%

ΔmAP vs GMR: +2.17 pp (vs GEQ: +1.90 pp) · ΔG1 mean vs GMR (.4/.5/.6): +1.28 pp · ΔG3 mean vs GMR (.4/.5/.6): +0.70 pp

BAM-DETR · G-mIoU@1



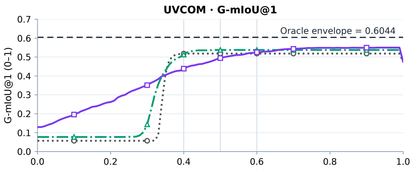
BAM-DETR · G-mIoU@3



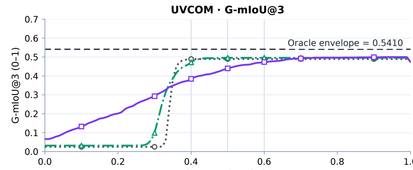
GMR baseline | mAP 23.25% · GMR + GEQ Full | mAP 23.00% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 23.99%

ΔmAP vs GMR: +0.74 pp (vs GEQ: +0.99 pp) · ΔG1 mean vs GMR (.4/.5/.6): +0.55 pp · ΔG3 mean vs GMR (.4/.5/.6): +1.09 pp

UVCOM · G-mIoU@1



UVCOM · G-mIoU@3



GMR baseline | mAP 7.12% · GMR + GEQ Full | mAP 8.61% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 17.94%

ΔmAP vs GMR: +10.82 pp (vs GEQ: +9.33 pp) · ΔG1 mean vs GMR (.4/.5/.6): -3.23 pp · ΔG3 mean vs GMR (.4/.5/.6): -5.67 pp

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E6) + GEQ − GMR baseline.

Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

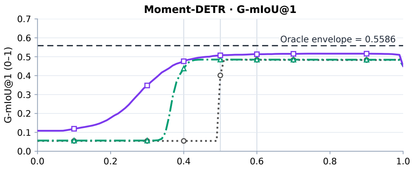
Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.

SoccerGMR Multi-Backbone Evaluation · GMR Baseline vs. GMR + GEQ vs. SI-Bind (E6) + GEQ

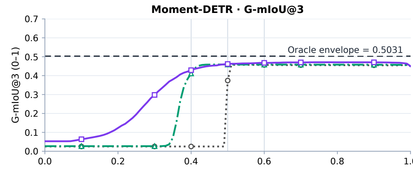
Standard-trained, fixed Val-selected best checkpoints → SoccerGMR Full-Decodable Test (2,970 queries)

Head-to-head threshold sensitivity comparison: G-mIoU@1, G-mIoU@3 and Oracle Presence Envelopes across τ ∈ [0.00, 1.00]

Moment-DETR · G-mIoU@1



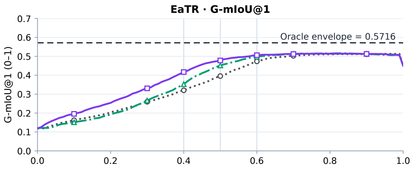
Moment-DETR · G-mIoU@3



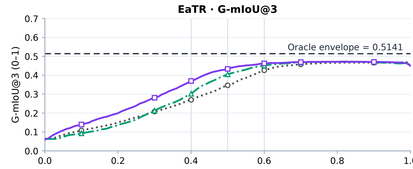
GMR baseline | mAP 5.47% · GMR + GEQ Full | mAP 5.47% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 12.95%

ΔmAP vs GMR: +7.48 pp (vs GEQ: +7.48 pp) · ΔG1 mean vs GMR (.4/.5/.6): +18.56 pp · ΔG3 mean vs GMR (.4/.5/.6): +16.74 pp

EaTR · G-mIoU@1



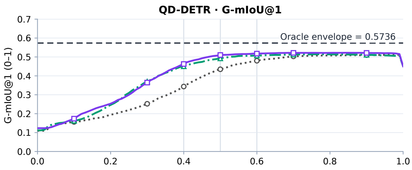
EaTR · G-mIoU@3



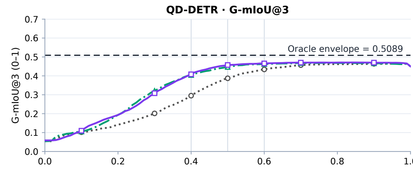
GMR baseline | mAP 15.19% · GMR + GEQ Full | mAP 15.26% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 14.36%

ΔmAP vs GMR: -0.83 pp (vs GEQ: -0.90 pp) · ΔG1 mean vs GMR (.4/.5/.6): +7.07 pp · ΔG3 mean vs GMR (.4/.5/.6): +7.43 pp

QD-DETR · G-mIoU@1



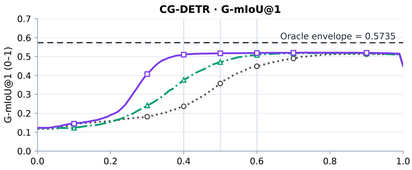
QD-DETR · G-mIoU@3



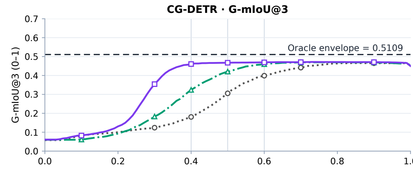
GMR baseline | mAP 12.65% · GMR + GEQ Full | mAP 13.22% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 14.98%

ΔmAP vs GMR: +2.33 pp (vs GEQ: +1.76 pp) · ΔG1 mean vs GMR (.4/.5/.6): +7.75 pp · ΔG3 mean vs GMR (.4/.5/.6): +7.17 pp

CG-DETR · G-mIoU@1



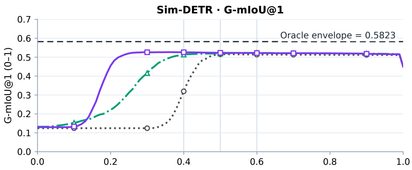
CG-DETR · G-mIoU@3



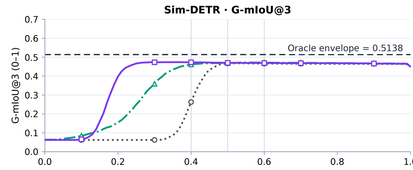
GMR baseline | mAP 14.67% · GMR + GEQ Full | mAP 14.85% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 15.47%

ΔmAP vs GMR: +0.80 pp (vs GEQ: +0.62 pp) · ΔG1 mean vs GMR (.4/.5/.6): +16.77 pp · ΔG3 mean vs GMR (.4/.5/.6): +17.07 pp

Sim-DETR · G-mIoU@1



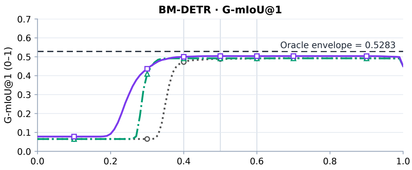
Sim-DETR · G-mIoU@3



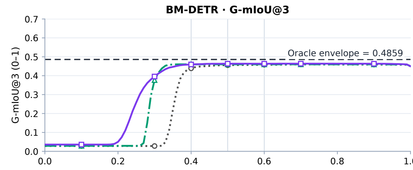
GMR baseline | mAP 19.18% · GMR + GEQ Full | mAP 20.58% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 19.96%

ΔmAP vs GMR: +0.78 pp (vs GEQ: -0.62 pp) · ΔG1 mean vs GMR (.4/.5/.6): +7.44 pp · ΔG3 mean vs GMR (.4/.5/.6): +7.23 pp

BM-DETR · G-mIoU@1



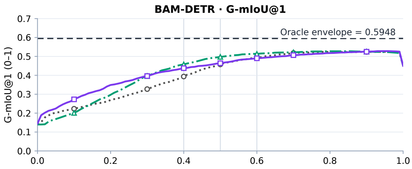
BM-DETR · G-mIoU@3



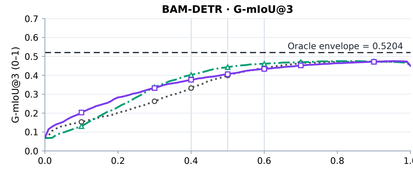
GMR baseline | mAP 3.65% · GMR + GEQ Full | mAP 4.32% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 5.63%

ΔmAP vs GMR: +1.98 pp (vs GEQ: +1.31 pp) · ΔG1 mean vs GMR (.4/.5/.6): +1.85 pp · ΔG3 mean vs GMR (.4/.5/.6): +1.13 pp

BAM-DETR · G-mIoU@1



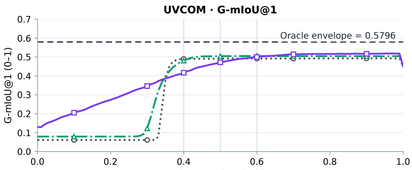
BAM-DETR · G-mIoU@3



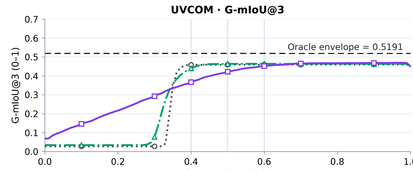
GMR baseline | mAP 20.62% · GMR + GEQ Full | mAP 20.26% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 21.56%

ΔmAP vs GMR: +0.94 pp (vs GEQ: +1.30 pp) · ΔG1 mean vs GMR (.4/.5/.6): +1.59 pp · ΔG3 mean vs GMR (.4/.5/.6): +1.58 pp

UVCOM · G-mIoU@1



UVCOM · G-mIoU@3



GMR baseline | mAP 6.40% · GMR + GEQ Full | mAP 8.13% · GMR + SI-Bind (E6) + GEQ (Ours) | mAP 16.55%

ΔmAP vs GMR: +10.15 pp (vs GEQ: +8.42 pp) · ΔG1 mean vs GMR (.4/.5/.6): -2.81 pp · ΔG3 mean vs GMR (.4/.5/.6): -4.55 pp

Curves: 101 fixed existence thresholds (0.00–1.00). Vertical guides indicate τ = 0.4, 0.5, 0.6. Δ: GMR + SI-Bind (E6) + GEQ − GMR baseline.

Oracle envelope: GT presence with each method's fixed candidate windows; GT-assisted ceiling diagnostic.

Evaluated without Test tuning or Test checkpoint selection. Native family ranking and duration scaling preserved.